

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Pharm. Examination

University Theory Examination December 2016

PH114/PH123 PHARMACEUTICAL CHEMISTRY-I

Date: 23.12.16, Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1: Attempt all questions. Each question is of one mark.

20

1. The term Slightly Soluble means _____ part of solvent require to dissolve 1 part of solute.

- [A] 1 to 10
- [B] 10 to 30
- [C] 30 to 100
- [D] 100 to 1000

2. In _____ limit test, we are comparing a color instead of turbidity.

- [A] Chloride
- [B] Lead
- [C] Iron
- [D] Sulphate

3. One of the following is sedative expectorant

- [A] NH_4Cl
- [B] Anise
- [C] Lemon
- [D] Eucalyptus

4. Radiopharmaceutical used in study of thyroid uptake

- [A] K-42
- [B] Cr-51
- [C] I-131
- [D] S-35

5. As per B.P which of the reagent is used for limit test of Sulphate?

- [A] Sodium Sulphate
- [B] Magnesium Sulphate
- [C] Barium Sulphate
- [D] None

6. CO₂ Cylinder is painted with _____ color.

- [A] Black
- [B] Red
- [C] Blue
- [D] Grey

7. _____ inorganic compound is used as an Opacifier.

- [A] Titanium Dioxide
- [B] Calamine
- [C] Zinc Oxide
- [D] Talc

8. _____ is an example of chemical antidote.

- [A] Copper Sulphate
- [B] Activated Charcoal
- [C] Kaolin
- [D] Magnesium Sulphate

9. In Povidone Iodine Solution _____ % of Iodine is available.

- [A] 20
- [B] 10
- [C] 15
- [D] 30

10. _____ Law of photochemistry is qualitative in nature.

- [A] Stark-Einstein Law
- [B] Beer's Law
- [C] Lambert's Law
- [D] Grothus-Draper Law

11. At constant temperature for the given mass of an ideal gas

- [A] The ratio of pressure and volume always remain constant
- [B] Volume always remain constant
- [C] Pressure always remain constant
- [D] Product of pressure and volume always remain constant

12. Following transition is responsible for phosphorescence.

- [A] Singlet to singlet
- [B] Singlet to triplet
- [C] Triplet to triplet
- [D] Triplet to singlet

13. Increase in _____ of the adsorbent increases the total amount of the gas adsorbed

- [A] Density
- [B] Volume
- [C] Surface area
- [D] Surface tension

14. A colligative property depends upon

- [A] Chemical nature of the particles
- [B] Size of the particles
- [C] Number of particles
- [D] Temperature of the solution

15. The reciprocal of viscosity is called

- [A] Surface tension
- [B] Frictional resistance
- [C] Fluidity
- [D] None of these

16. Which is not true about thermodynamics?

- [A] It ignores the internal structure of atoms and molecules
- [B] It involves the matter in bulk
- [C] It is concerned only with the initial and final states of the system
- [D] It is not applicable to macroscopic systems

17. ORS (recommended by WHO) contain NaCl in amount of

- [A] 3.5gm
- [B] 1.3gm
- [C] 2.5gm
- [D] 4.5gm

18. Which of the following topical agent is oily in nature?

- [A] Talc
- [B] Calamine
- [C] Dimethicone
- [D] Titanium dioxide

19. Which of the following inorganic compound is used as dental cement?

- [A] ZnCl_2
- [B] NaCl
- [C] KMnO_4
- [D] ZnO

20. Milk of magnesia is

- [A] Hydrated Magnesium Silicate
- [B] Hydrated Magnesium Oxide
- [C] Hydrated Magnesium Hydroxide
- [D] Hydrated Magnesium Carbonate

Q 2	Attempt any <u>TWO</u> of the following	A
	Define a following terminologies (Any Five)	05
	I. Achlorhydria	
	II. Astringent	
	III. Antacid	
	IV. Protective and Adsorbents	
	V. Cathartics	
	VI. Expectorants	
B	Write a principle for the limit test of Chloride and Sulphate.	05
C	What is monograph? Write the content of monograph.	05
Q 3	Attempt any <u>TWO</u> of the following	
A	Classify expectorants and write a short note on Ammonium Chloride as an expectorant.	05
B	Compare Real gas and Ideal gas.	05
C	Draw and describe Jabolanski's diagramme.	05
SECTION – II		
Q 4	Attempt any <u>FOUR</u> of the following	
A	What is nuclear chemistry? Explain various radia active rays.	05
B	Explain nuclear fission reaction with suitable example.	05
C	Write a note on 1 st and 2 nd law of thermodynamics.	05
D	What is adsorption? Explain types of adsorption with their mechanism.	05
E	Explain Dalton's Law.	05
F	What do you mean by colligative property? Enumerate various colligative properties of liquids and explain any one detail.	05
Q 5	Attempt any <u>FOUR</u> of the following	
A	Explain laws of photochemistry.	05
B	What is Cyanide Poisoning? Explain its treatment.	05
C	Give the preparation, assay and uses of Sodium Bicarbonate.	05
D	Write the preparation, assay and uses of Boric Acid	05
E	Explain the preparation and assay of Copper Sulphate.	05
F	Give the preparation, assay and uses of Magnesium Carbonate.	05